

Customer Churn Analysis & Customer Intelligence

A validated retention analytics case study

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Portfolio project | Data preparation, churn diagnostics, customer intelligence

This report audits the supplied SQLite database, cleaned customer-level export, analytical notebook, and existing visualizations. It preserves the source findings where valid, corrects denominator and interpretation issues, and translates the evidence into practical retention priorities.

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1 Executive Summary

This project analyzes a 21-customer subscription dataset assembled from three SQLite tables: customer attributes, subscription records, and support interactions. The objective is to understand who has churned, which customer segments show elevated observed churn, what operational signals accompany churn, and where retention intervention should be prioritized.

The validated analysis finds that 6 of 21 customers have a recorded cancellation, producing an observed churn rate of **28.57%** and a retention rate of **71.43%**. The churned accounts represent **73.94** in the dataset's monthly-charge units, or **18.68%** of the portfolio's total monthly charges of **395.79**. Basic-plan churn is **60.00%** versus **22.22%** for Standard and **14.29%** for Premium. Monthly contracts show **55.56%** churn versus **8.33%** for annual contracts.

The most concentrated acquisition risk is the Referral segment: **5 of 6** Referral customers churned, compared with **1 of 6** Paid customers and **0 of 9** Organic customers. Geography is directionally useful but highly sample-constrained: Karnataka records 2 of 2 customers churning, Meghalaya 2 of 3, and Telangana 1 of 2. These rates should be treated as prioritization signals rather than stable market benchmarks.

The strongest descriptive signal is the pre-existing churn score. All 6 customers classified as high risk under the notebook's thresholds churned, while none of the 13 low-risk or 2 medium-risk customers churned. This is a useful validation of the score's alignment with observed cancellations, but it is **not** an out-of-sample model-performance result because the risk band is derived directly from churn score and the analysis uses the same small sample to define and inspect the relationship.

28.57% \ Observed churn rate

73.94 \ Monthly charges at risk

55.56% \ Monthly-contract churn

The business implication is clear: retention effort should begin with **high-score, monthly-contract, Basic-plan, and Referral-acquired customers**, while support and experience teams should use escalation and low CSAT signals as intervention triggers. The evidence supports targeted retention experiments and better measurement discipline—not blanket discounts or causal claims.

2 Business Problem & Objectives

Customer churn reduces recurring revenue, weakens lifetime value, and makes acquisition spending less efficient. A useful churn review therefore needs to do more than count cancellations: it should identify concentration by product, contract, acquisition source, geography, and customer experience; distinguish descriptive association from causal evidence; and define an operational sequence for action.

This project addresses five questions:

1. What is the observed churn and retention position of the customer portfolio?
2. Which plan, contract, acquisition, and geographic segments have the highest observed churn?
3. How do churn score, tenure, charges, CLTV, support, escalations, and CSAT relate to churn?
4. What are the documented reasons for cancellation, and how much monthly-charge exposure do they represent?
5. Which retention actions are most directly supported by the evidence, and what measurement improvements are required next?

3 Dataset & Analytical Scope

The supplied SQLite database contains three tables joined on `customerid`. The final analysis population is the subscription table at one row per customer, left-joined to standardized customer attributes and a customer-level support summary.

Source table	Rows	Role	Important fields used
db_customer	21	Customer profile	customerid, country, state, gender, dob
db_subscription	21	Subscription and outcome	dates, subscription type, plan, contract, charges, CLTV, churn score, cancellation reason
db_support	9	Support events	customerid, complaint date, escalations, CSAT score
Final merged analysis	21	One row per customer	subscription base plus standardized attributes and latest support record

The primary deliverable is a descriptive business analysis of this supplied snapshot. It is not a production churn model, causal study, or statistically powered market comparison. Monetary values are reported in the units present in the source field `monthly_charges`; the data does not provide a currency label, so the report does not assume one.

4 Data Preparation & Quality Validation

The notebook's cleaning logic was reproduced and tested against the database and exported CSV. Customer IDs are unique in the customer and subscription tables, and the final merged dataset contains 21 unique customer records. The subscription-based churn flag correctly matches the presence of a cancellation date for all 21 records.

Customer preparation renamed `name` to `customer_name`, removed the unused interests and completely empty `pincode` fields, converted `dob` to a date, standardized Men and Women to Male and Female, and filled three missing country values from the observed state-to-country mapping. The result contains no missing customer-level profile fields used in the analysis.

Subscription preparation converted `subscription`, `renewal`, and `cancellation` dates to date fields. Churn was defined as `cancellation_date` not null. Support preparation converted complaint dates, removed the empty auxiliary column and comments from the analytical table, counted support events by customer, and retained the latest support record per customer to prevent the merge from duplicating the subscription base.

Several improvements were applied to the interpretation layer. The source spelling `Referral` is displayed as **Referral** in this report without changing the underlying data. Support metrics are reported with explicit denominators because only 7 of 21 customers have support records. Tenure is labeled as measured through the report date, 25 August 2026, because the notebook used the current date for active customers; this makes the result reproducible for this report but time-dependent for future reruns.

Validation note. The notebook's escalation rate of 19.05% uses all 21 customers as the denominator, treating customers without a support record as non-escalated. The more decision-useful support-case rate is 4 of 7 supported customers, or **57.14%**. Both values are retained and labeled distinctly in this report.

5 Feature Engineering & Methodology

The analysis constructs a compact customer-level analytical frame. The derived fields are summarized below.

Feature	Definition	Analytical use
churn_flag	1 when cancellation date exists, otherwise 0	Outcome measure for churn and retention rates
complaint_count	Count of support rows associated with each customer before latest-record deduplication	Support-contact intensity and service burden
tenure_days	Cancellation date minus start date for churned customers; report date minus start date for active customers	Lifecycle context; descriptive only
escalations_flag	Y mapped to 1 and N mapped to 0	Support escalation association
churn_risk	low: score < 50; med: 50–69; high: score >= 70	Operational prioritization band derived from churn score
cancellation_month	Month extracted from cancellation date	Monthly churn timing

The workflow is: **Raw SQLite tables** → **data cleaning** → **table integration** → **customer-level feature engineering** → **KPI validation** → **segment analysis** → **visual diagnostics** → **business implications** → **recommendations**. Descriptive percentages use customer counts within each segment. Correlations are Pearson correlations calculated on pairwise complete observations; the available pair count is 21 for the core subscription variables and 7 for support variables.

6 Portfolio KPIs

KPI	Value	Interpretation
Customer records	21	Small analytical snapshot; findings are directional and should be validated on a larger history.
Churned customers	6	Customers with a recorded cancellation date.
Observed churn rate	28.57%	6 churned customers divided by 21 total customers.

KPI	Value	Interpretation
Retention rate	71.43%	15 customers without a recorded cancellation date.
Average monthly charges per customer	18.85	Mean of the source monthly_charges field; currency not specified.
Monthly charges at risk	73.94	Sum of monthly charges for the 6 churned customers.
Average tenure	1,524 days / 4.17 years	Observed duration through 25 August 2026 for active customers; cancellation date for churned customers.
Support coverage	$7 / 21 = 33.33\%$	Customers with at least one support record.
Escalation rate among support cases	$4 / 7 = 57.14\%$	Correct denominator for customers represented in the support table.
Average complaints per customer	0.43	9 support events divided by 21 customers; among supported customers, the average is 1.29.

7 Segment Findings

7.1 Plan and contract structure

Basic is the highest-churn plan: 3 of 5 customers churned, or **60.00%**. Standard is 2 of 9, or **22.22%**, and Premium is 1 of 7, or **14.29%**. The Basic plan therefore combines the highest observed churn with the lowest average monthly charge of the three plan groups. This is an important targeting signal, but the sample is too small to infer whether price, product experience, customer mix, or another factor drives the pattern.

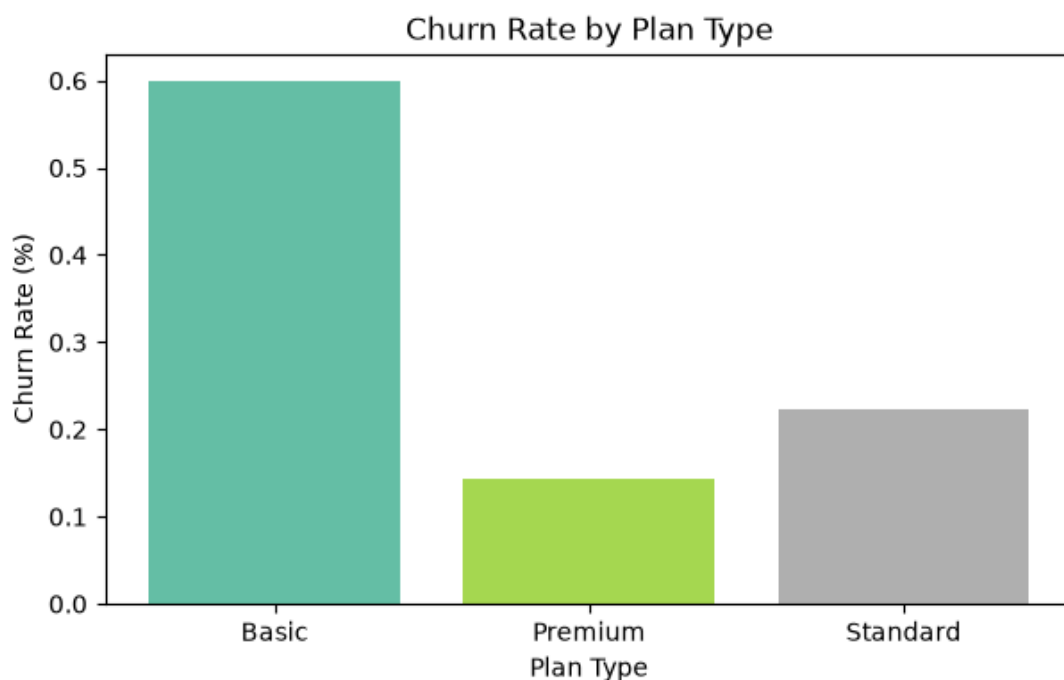


Figure 1: Existing churn-rate-by-plan visualization. The source plot stores rates as proportions; the validated percentages are Basic 60.00%, Premium 14.29%, and Standard 22.22%.

Plan	Customers	Churned	Churn rate	Monthly charges
Basic	5	3	60.00%	52.95
Standard	9	2	22.22%	123.91
Premium	7	1	14.29%	218.93

Contract type is a sharper differentiator than plan type in this sample. Monthly contracts show 5 of 9 customers churned, or **55.56%**. Annual contracts show 1 of 12, or **8.33%**. This pattern supports lifecycle interventions around monthly renewal or early-tenure moments, but it should not be read as proof that changing contract type alone will prevent churn.

7.2 Acquisition source

Referral customers show the highest observed churn: 5 of 6, or **83.33%**. Paid acquisition shows 1 of 6, or **16.67%**, while Organic shows 0 of 9, or **0.00%**. Referral also contributes only 74.94 in monthly charges because five of its six customers have churned. The pattern warrants a source-quality review: confirm referral expectations, onboarding quality, offer fit, and whether referred customers are being attracted into higher-risk plans or monthly contracts.

Acquisition source	Customers	Churned	Churn rate	Monthly charges
Organic	9	0	0.00%	145.91
Paid	6	1	16.67%	174.94
Referral	6	5	83.33%	74.94

7.3 Geography

The state-level view identifies Karnataka, Meghalaya, and Telangana as the highest observed-rate groups. Karnataka is 2 of 2 churned, Meghalaya is 2 of 3, and Telangana is 1 of 2. Delhi is 1 of 4, or 25.00%. The remaining displayed groups have no recorded churn in this snapshot.

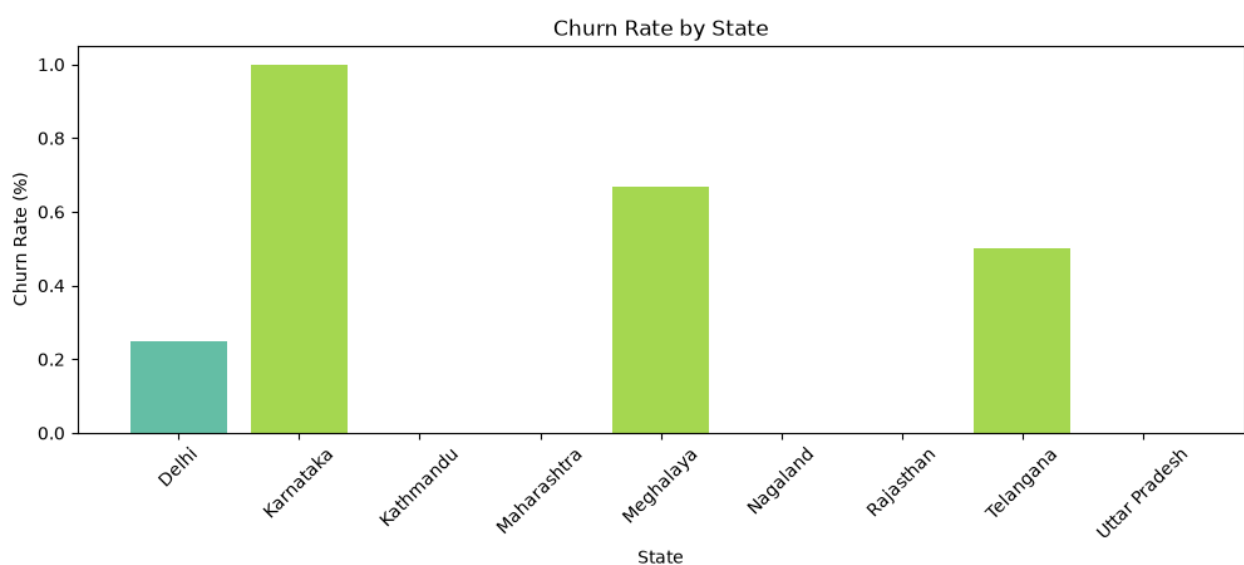


Figure 2: Existing state-level churn visualization. Rates are shown with denominators in the accompanying interpretation because several states contain only one or two customers.

The geography field also contains Kathmandu as a state-like value, with Nepal as the country for those customers, while the other states are Indian locations. This is not a blocker for the descriptive analysis, but production reporting should separate country and state/province dimensions cleanly and maintain a validated geography hierarchy.

7.4 Churn reasons and timing

Among the 6 churned customers, **Switched to competitor** is the only repeated cancellation reason, accounting for 2 customers, or 33.33% of churn. The remaining four reasons occur once each: Forgot to cancel trial, Not enough content, Poor streaming quality, and Too expensive. The evidence suggests at least three retention workstreams—competitive differentiation, trial conversion/renewal reminders, and product/content/service experience—rather than a single universal offer.

Cancellation reason	Customers	Share of churn	Monthly charges
Switched to competitor	2	33.33%	25.98
Forgot to cancel trial	1	16.67%	16.99
Not enough content	1	16.67%	8.99
Poor streaming quality	1	16.67%	7.99
Too expensive	1	16.67%	13.99

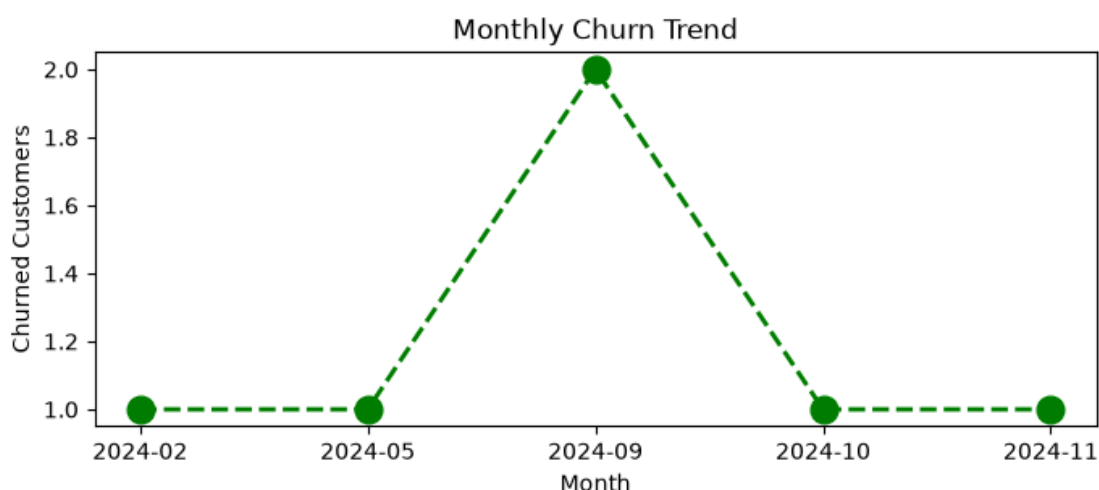


Figure 3: Existing monthly churn trend. The six churn events occurred across five cancellation months, with a peak of two in September 2024.

7.5 Customer experience signals

Support data is sparse but concentrated among churned customers: all 6 churned customers have a support record, while only 1 retained customer does. Within the support-covered population, churned customers have average CSAT of **34.17** and a 66.67% escalation rate, compared with average CSAT of **90.00** and 0.00% escalation among the one retained supported customer. These patterns are directionally important, but the retained comparison has only one record and cannot support stable inference.

The operational recommendation is to treat low CSAT, an escalation, or repeated support contact as a trigger for human review—without assuming that support causes churn. The data may instead capture customers who were already dissatisfied or already considering cancellation.

8 Churn Score, Risk Bands & Correlation

The notebook's risk bands produce a striking descriptive split. All 6 high-risk customers churned, while none of the 13 low-risk or 2 medium-risk customers churned. The high-risk group carries 73.94 in monthly charges at risk. Because churn_risk is mechanically derived from churn_score, and the score is itself strongly aligned with recorded churn, this result validates alignment with the observed target but does not establish independent predictive performance.

Risk band	Customers	Churned	Observed rate	Monthly charges
Low: score < 50	13	0	0.00%	286.87
Medium: 50–69	2	0	0.00%	34.98
High: score >= 70	6	6	100.00%	73.94

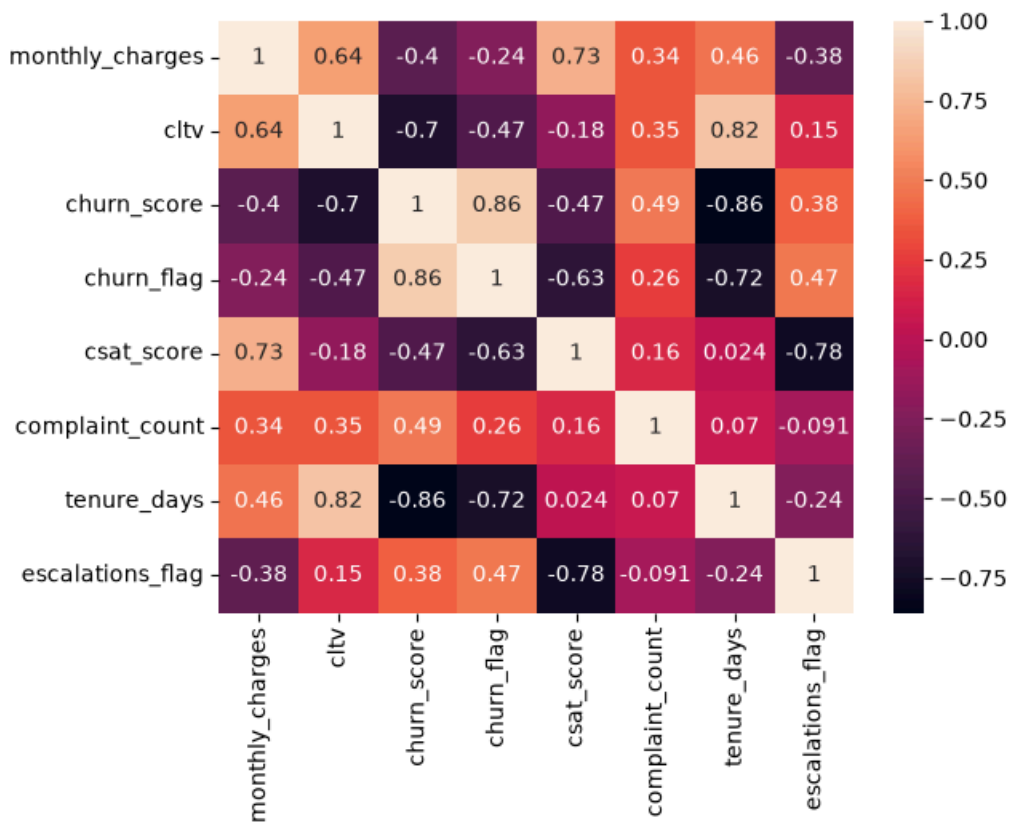


Figure 4: Pairwise Pearson correlations from the core and support-enriched customer frame. Support variables use 7 complete observations; most subscription variables use 21.

The most relevant relationships with the churn flag are:

Variable compared with churn flag	r	Interpretation
Churn score	+0.8572	Strong positive association; expected because the score is designed to reflect churn propensity.
Tenure days	-0.7232	Longer observed tenure is associated with lower churn in this snapshot.
CSAT score	-0.6319	Lower satisfaction is associated with churn; based on 7 support-covered customers only.

Variable compared with churn flag	r	Interpretation
CLTV	-0.4673	Moderate negative association; may reflect short-tenure or low-value churned accounts.
Escalation flag	+0.4714	Moderate positive association; based on 7 support-covered customers only.
Monthly charges	-0.2381	Weak negative association; not a reliable pricing conclusion.

Correlation does not prove causation. In particular, the score-to-churn and risk-to-churn relationships include derived or pre-existing risk information, while the support correlations have a small and selected denominator. The encoded heatmap and pairplot supplied with the project are therefore treated as methodological diagnostics rather than primary business evidence: ordinal encoding of nominal categories can create misleading numeric relationships, and the pairplot is too dense to improve executive interpretation.

9 Priority Insights

Priority	Observed pattern	Business interpretation	Evidence strength
1	All 6 high-risk customers churned	Use the score as a triage queue, then test whether interventions reduce subsequent cancellations.	High within sample; not out-of-sample
2	Monthly contracts churn at 55.56% vs 8.33% annual	Monthly renewal moments are the clearest contract-level intervention point.	Strong descriptive
3	Referral churn is 83.33% vs Organic 0.00%	Acquisition-source quality and onboarding fit require review.	Strong but small n
4	Basic churn is 60.00% vs Premium 14.29%	Basic customers should receive product-value and content-fit diagnostics.	Strong but small n
5	Churn reasons are fragmented; competitor switching repeats twice	Retention messaging should combine competitive value, trial conversion, and service-quality actions.	Moderate
6	Churned support cases have lower CSAT and more escalations	Escalation and low CSAT can be used as service-recovery triggers.	Directional; n=7

10 Business Recommendations

10.1 Establish a high-risk retention queue

Finding → implication → action. All 6 customers in the high-risk band churned, and the band accounts for 73.94 in monthly charges at risk. Create a weekly queue for high-score customers, combining a service check, product-value message, and contract/renewal review. Track contact, offer, response, and subsequent outcome so the score becomes an intervention workflow rather than a static chart.

10.2 Focus renewal intervention on monthly contracts

Monthly contracts churn at 55.56%, compared with 8.33% for annual contracts. Introduce a renewal calendar for monthly customers, with value reminders, frictionless plan review, and cancellation-intent capture before the next renewal. Any upgrade or incentive should be tested against a control group and evaluated on retained customers and net monthly-charge impact, not churn alone.

10.3 Audit the Referral acquisition journey

Five of six Referral customers churned. Review referral messaging, expectations at sign-up, onboarding completion, initial product usage, and plan/contract mix. Add a 30-day check-in for new referred customers and compare their activation and early-support behavior with Organic and Paid cohorts. Because the count is six, the first objective is diagnosis and measurement—not immediately suspending the channel.

10.4 Improve Basic-plan value realization

Basic has the highest observed churn at 60.00% and the lowest average monthly charge. Interview churned Basic customers, analyze the recorded reasons, and test content discovery, streaming-quality, and plan-value interventions. Monitor whether improvements reduce cancellations without shifting customers into lower-value behavior or increasing service cost disproportionately.

10.5 Operationalize service-recovery triggers

All churned customers in the support-enriched dataset have a support record; their average CSAT is 34.17 and escalation rate is 66.67%. Route low-CSAT and escalated cases to a service-recovery playbook with a defined response time, ownership, and closure reason. Expand support capture for retained customers so the signal can be evaluated without selection bias.

10.6 Strengthen cancellation-reason capture and measurement

The most common reason is switching to a competitor, while four other reasons occur once each. Make cancellation reason mandatory and mutually exclusive where appropriate, add a structured secondary reason, and retain the customer's plan, contract, acquisition source, and support history at cancellation. This will allow future analysis to separate competitive, product, price, and process drivers.

11 Measurement Roadmap & Limitations

The current analysis is valuable as a compact portfolio case study and an initial retention diagnostic, but it has material limitations. The sample contains only 21 customers and 6 churn events, so rates can move by 16.67 percentage points with a single customer. State-level rates are especially unstable. Support fields are present for only 7 customers, and the latest-record deduplication intentionally simplifies the support history rather than preserving every event as a longitudinal fact table.

The data does not include a full observation window, customer acquisition dates beyond subscription start, product usage, cohort definitions, experiment assignments, or a currency label for charges. The notebook's active-customer tenure is time-dependent because it uses the current date. The risk bands are derived from churn score, and churn score is strongly aligned with the churn flag; these variables must not be used as independent predictors without leakage controls.

A stronger next iteration should: (1) collect at least 12–24 months of customer-level snapshots; (2) define a fixed observation date and prediction horizon; (3) separate feature creation time from churn outcome time; (4) retain all support events in a one-to-many event table; (5) evaluate models with time-based validation, calibration, precision/recall, and lift; and (6) run controlled retention experiments with net revenue and customer-experience guardrails.

12 Conclusion

The validated dataset shows an observed churn rate of 28.57% across 21 customers, with 73.94 in monthly charges associated with churned accounts. Churn is concentrated in high-risk scores, monthly contracts, Referral acquisition, and the Basic plan. Support-enriched churned customers also show low CSAT and more escalations, although that evidence is limited by sparse coverage and selection.

The immediate business focus should be targeted intervention rather than broad-based discounting: prioritize high-score monthly and Basic customers, audit Referral onboarding and channel fit, and use service-recovery triggers for low-CSAT or escalated cases. The most important analytical next step is to improve the data foundation so future retention decisions can be evaluated over time and against credible controls.

13 Technical Stack

The supplied notebook verifies the use of Python with Pandas, NumPy, SQLite, Matplotlib, and Seaborn. SQLite is used for source-table storage and extraction; Pandas and NumPy support cleaning, joining, feature engineering, aggregation, and validation; Matplotlib and Seaborn support the supplied visualizations.

14 Source Materials

This report was prepared strictly from the supplied project artifacts: `customer_churn.db`, `exported_churn_data.csv`, `churn_analysis.ipynb`, `ChurnAnalysisandCustomerIntelligence.pdf`, and the provided PNG visualizations. The database and notebook were treated as primary sources; the exported CSV and existing figures were used as cross-checks and exhibits. All numerical values in this report are recomputed from the supplied data or transcribed from validated notebook outputs.

Portfolio note. This report is intentionally transparent about sample size, denominator choices, derived variables, and uncertainty. That distinction between a compelling descriptive signal and a production-ready prediction is central to responsible business analytics.